

Harini Ballalasamudra Mohankumar

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Education

Syracuse University - College of Engineering & Computer Science

Syracuse, New York

Master of Science in Computer Science, Relevant coursework:

January 2023 – Present

Computer Architecture, Deep Learning and Reinforcement Learning AI, NLP, Big Data, Data Mining

Visvesvaraya Technological University - BMS College of Engineering

Bengaluru, Karnataka

Bachelor of Engineering in Computer Science, Relevant coursework:

August 2016 – June 2020

Big Data and Analytics, Databases (relational and non-relational), Cloud Computing, Python, Java, Computer Networks, Algorithms, Operating Systems

Technical Skills

Languages: Python, JavaScript

Frameworks: FastAPI, Flask

Databases: SQL (PostgreSQL), NoSQL (MongoDB)

Tools: Git, Docker, Kubernetes

Experience

iConsult Collaborative at Syracuse University

Syracuse, New York, United States (Hybrid)

Data Engineer / Python Developer

Mar 2024 – December 2024

- Developed robust REST APIs using **FastAPI** to fetch and process real-time health data from Garmin wearables and internal applications for patient analysis.
- Leveraged asynchronous programming techniques to enhance API performance and scalability for handling high-volume data streams.
- Collaborated with cross-functional teams to seamlessly integrate backend services with frontend components, ensuring secure and efficient data flow.

Cermati

Bengaluru, Karnataka

Data Engineer

January 2020 – August 2022

- Built scalable **ETL** data ingestion pipelines with **Python**, **PostgreSQL**, **BigQuery**, and **Apache Kafka**, enhancing data processing throughput.
- Optimized **SQL queries** and database designs, improving query execution times by 40% and enabling seamless scalability for large datasets in **PostgreSQL** and **BigQuery**.
- Designed and implemented backend systems for credit scoring of loan applications using **Python**, optimizing real-time data processing and reducing computation time by 30%, while collaborating with the risk analyst team to ensure accuracy and compliance with scoring models.
- Built high-performance **RESTful APIs** using **Flask** for internal event tracking of customer activities.
- Designed and deployed containerized solutions using **Docker** and **Kubernetes**, ensuring consistent production environments and reducing deployment issues by 20%.
- Proactively monitored production environments using **Prometheus** and **Grafana**, achieving 99% uptime by identifying and resolving issues before impact.
- Administered Linux-based servers, optimized performance, and automated system tasks using **Bash** and **Ansible**.

Projects

Rental Management Web Application

- Developed a scalable backend service with **FastAPI** in **Python** to support comprehensive rental item management.
- Employed asynchronous programming to efficiently handle multiple simultaneous requests for tracking rentals and managing inventory.
- Partnered with frontend developers to integrate a dynamic **ReactJS** interface, providing a smooth and interactive user experience.
- Streamlined deployment processes by containerizing the application with **Docker** and orchestrating its rollout using **Kubernetes**, enhancing reliability and uptime.